AI Cyber Threat Detection and Auto-Respond System

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***Abstract*—In today's digital world, security is becoming more important with the increase in online systems. This project presents a simple yet effective AI-based cybersecurity system that detects suspicious login attempts and responds automatically.**

The system uses a web-based login interface and a webhook-based backend to process user activity. It combines rule-based logic (using IF conditions) and artificial intelligence to make smart decisions. OpenAI analyzes the login behavior and classifies it as either normal or suspicious.

If the system detects a high-risk situation, it automatically sends alerts through Email and Telegram. At the same time, all activities are stored in Google Sheets for record keeping and future analysis. The system also sends a real-time response back to the user interface.

This project demonstrates how modern tools like AI and workflow automation can be used together to create a smart, responsive, and practical cybersecurity solution.

***Index Terms*—Cybersecurity, Artificial Intelligence, Automation, Webhook, Threat Detection, n8n**

I. Introduction

With the rapid growth of online platforms, cybersecurity has become a major concern. Almost every application today uses a login system, which acts as the first level of protection. However, these systems are often targeted by attackers who try to gain unauthorized access.

One of the most common attacks is a brute-force attack, where an attacker repeatedly tries different passwords until the correct one is found. If there is no proper detection system, these attacks can continue without any restriction.

This project focuses on solving this issue by building an intelligent system that can detect suspicious login behavior in real time. The system not only detects the problem but also responds immediately without human intervention.

The goal is to show how even a simple system can become smart and secure when AI and automation are used together.

II. Scope of the Project

The scope of this project is limited to detecting suspicious login attempts in a controlled environment. It is mainly designed for learning and demonstration purposes.

This system does not require complex infrastructure or advanced hardware. It shows how easily available tools can be integrated to create a functional cybersecurity solution.

Although the system is simple, the concept can be extended to real-world applications such as banking systems, websites, and mobile apps.

III. Problem Statement

Many existing login systems allow users to attempt login multiple times without any intelligent monitoring. This creates a serious security risk.

There is a need for a system that can:

- Monitor login behavior continuously
- Detect unusual patterns such as repeated failures
- Make smart decisions based on data
- Respond automatically without delay
- Notify administrators about potential threats

This project aims to address these problems using a combination of AI and automation tools.

IV. Objectives

The main objectives of this project are:

- To design a system that monitors login attempts
- To implement IF-based decision logic for initial filtering
- To use AI for analyzing user behavior
- To classify login attempts as normal or suspicious
- To trigger alerts only when necessary
- To store all data for tracking and analysis
- To provide real-time response to users

V. Methodology

A. Data Collection

The system starts with a simple HTML login page where users enter their credentials. It also tracks the number of failed login attempts.

B. Webhook Integration

The collected data is sent to an n8n webhook. This webhook acts as a bridge between the frontend and backend.

C. Decision Logic using IF Node

The system uses an IF node to check the number of failed attempts:

- If failed attempts are greater than 5, it is treated as suspicious
- Otherwise, it is considered normal

This step helps in early filtering before sending data to AI.

D. AI-Based Analysis

OpenAI is used to analyze the login data and generate structured output. It provides:

- Status (normal or suspicious)
- Threat type (e.g., brute-force)
- Threat level (low or high)
- Recommended action (allow or lock account)

E. Risk-Based Automation

Another IF node is used to check the threat level:

- If threat level is high, alerts are sent
- If threat level is low, no alerts are sent

F. Alerts and Notifications

In case of high risk:

- Email alert is sent
- Telegram message is triggered

G. Data Storage

All results are stored in Google Sheets, including:

- Username
- Failed attempts
- Status
- Threat level
- Action
- Time of login

H. Response to User

Finally, the system sends a response back to the login interface, informing whether login is allowed or blocked.

VI. Tools and Technologies

This project uses the following tools:

- HTML for frontend login page
- n8n for workflow automation
- OpenAI API for intelligent decision making
- Email service for alerts
- Telegram Bot API for notifications
- Google Sheets for storing data

VII. System Design

A. Frontend

A simple and user-friendly login page that collects user input and sends it to the backend.

B. Backend

The backend workflow is created using n8n and includes:

- Webhook node
- IF nodes for decision making
- OpenAI node
- Alert nodes
- Google Sheets node

C. Workflow

User → Webhook → IF (Attempts) → OpenAI → IF (Risk) → Alerts → Storage → Response

VIII. System Implementation

The system works by connecting different components step by step. When a user attempts to log in, the data is sent to the webhook. The workflow then processes the data using logic and AI.

The system checks for suspicious behavior, generates a response, and takes action accordingly. This integration shows how multiple tools can work together smoothly.

IX. Working of the System

- User enters login credentials
- System tracks failed attempts
- Data is sent to webhook
- IF node evaluates attempts
- OpenAI analyzes behavior
- IF node checks threat level
- Alerts are sent if needed
- Data is stored in Google Sheets
- Response is shown to user

X. Results

The system was tested under different conditions. The results were as follows:

- Normal login attempts were processed successfully
- Multiple failed attempts were correctly identified as suspicious
- Alerts were sent only in high-risk cases
- Data was accurately stored in Google Sheets

The system performed efficiently and gave correct outputs in real time.

XI. Discussion

This project shows that even a simple system can become powerful when AI and automation are combined. The use of IF logic improves decision making, while AI adds intelligence to the system.

Although the system is basic, it clearly demonstrates the core idea of smart cybersecurity systems.

XII. Future Work

There are many ways to improve this system:

- Use a database instead of Google Sheets
- Add advanced machine learning models
- Improve UI design
- Add multi-factor authentication
- Deploy the system on a live server

XIII. Conclusion

This project successfully demonstrates an AI-based cybersecurity system that detects suspicious login attempts and responds automatically.

The system is simple, effective, and easy to understand. It highlights the importance of using AI and automation in modern security systems.

References

[1] OpenAI Documentation
[2] n8n Workflow Automation Documentation
[3] OWASP Cybersecurity Guidelines